Glioblastoma Clinical Data Analysis Report

Summary of Clinical and Technical Patient Characteristics

Characteristic	Healthy_controls N = 42[1]	Metastatic N = 24[1]	Non-metastatic N = 18[1]	p-value[2]
age	48 (20)	48 (19)	48 (22)	>0.9
bmi	25.6 (6.4)	26.4 (6.2)	24.5 (6.7)	0.7
days_survived	906 (301)	739 (198)	1,129 (270)	<0.001
censored	16 / 42 (38%)	4 / 24 (17%)	12 / 18 (67%)	0.004
batch				0.8
B1	17 / 42 (40%)	10 / 24 (42%)	8 / 18 (44%)	
B2	12 / 42 (29%)	4 / 24 (17%)	4 / 18 (22%)	
B3	13 / 42 (31%)	10 / 24 (42%)	6 / 18 (33%)	
slide_area	127,168 (32,988)	51,976 (21,962)	48,716 (25,462)	<0.001

[1] Mean (SD); n / N (%)

[2] Kruskal-Wallis rank sum test; Pearson's Chi-squared test; Fisher's exact test

Clinical characteristics were broadly similar across the three groups. There was no significant difference in the age and BMI between groups ($p > 0.05$), suggesting the cohorts are well matched for these variables. Days survived differed significantly between groups ($p < 0.001$), reflecting the poorer prognosis of metastatic disease. Batch distribution was also similar across groups ($p = 0.8$), indicating that sequencing batch is unlikely to confound expression analyses. However, slide area differed substantially between groups ($p < 0.001$), with healthy samples having larger tissue regions sequenced. This difference may influence the number of detected transcripts and could introduce technical bias.

Principle Component Analysis

In the PCA plot coloured by batch, samples from the three batches are largely intermixed across the PC1-PC2, indicating no strong batch-driven clustering. While there is a slight tendency for B2 samples toward the centre, this pattern is relatively weak and not clearly separated. Interestingly, when examining the PC3-PC4 plot, this subtle batch structure becomes less apparent, suggesting that batch effects do not contribute strongly to the variation in the dataset.

In contrast, Slide Area shows a clearer pattern. In the PC1-PC2 plot, samples with high slide area tend to cluster toward the negative side of PC1, whereas samples with low slide area are more frequently located toward the positive side, indicating that this variable may influence the first principal component. Notably, this separation remains visible in the PC3-PC4 plot, suggesting that slide area contributes to multiple axes of variation in the data. This persistent structure indicates that slide area may represent a meaningful technical source of variability and should therefore be corrected for before downstream analyses to minimise potential confounding effects.

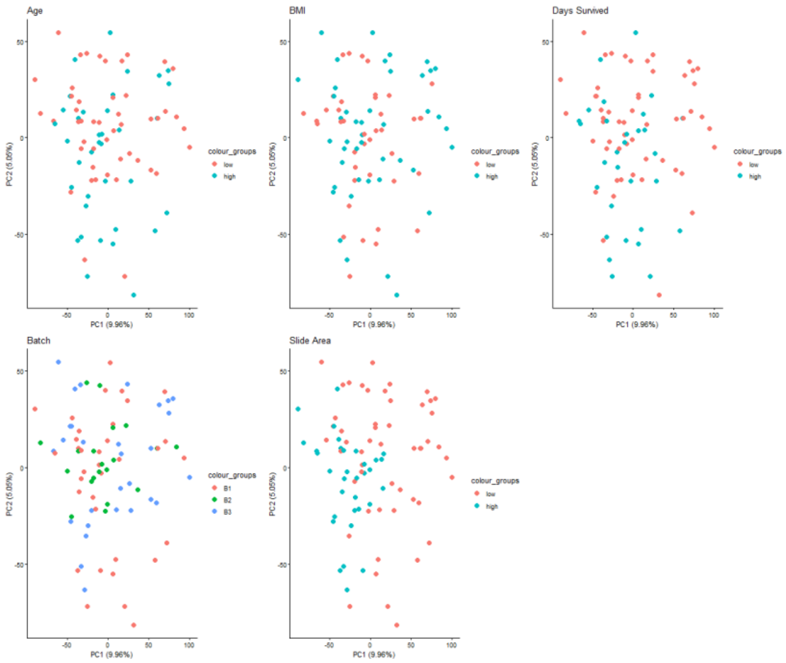

Figure 1: Principal Component Analysis (PCA) of variance across clinical and technical variables. The plots illustrate the distribution of samples along the first two principal components (PC1: 9.96%, PC2: 5.05%). Samples are coloured by status (low/high or B1/B2/B3).

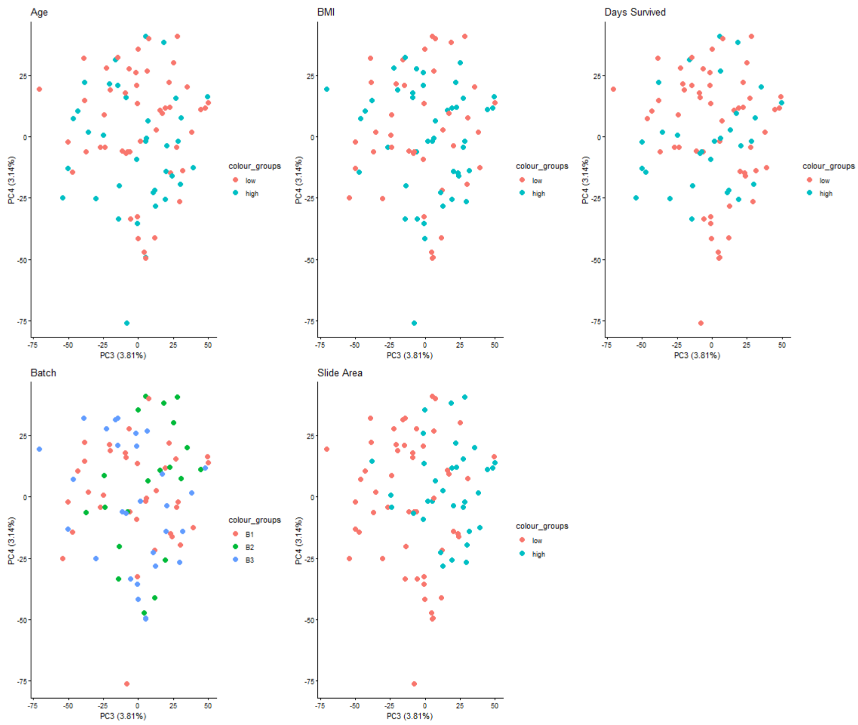

Figure 2: Principal Component Analysis (PCA) of variance across clinical and technical variables. The plots illustrate the distribution of samples along the first two principal components (PC3: 3.81%, PC4:3.14%). Samples are coloured by status (low/high or B1/B2/B3).